JONATHAN CARROLL, PHD

I provide software design and implementation for statistics, analysis, visualisation, exploration, curation, transfer, and storage - all things data.

I enjoy solving complex puzzles, both mathematical, algorithmic, and biological. I build tools to ease data flow between storage and scientists.

I am comfortable with a UNIX/Linux environment and a variety of platform ecosystems. I have successfully navigated remote positions since 2017.

EDUCATION

2009 | 2006

PhD
Theoretical Physics

University of Adelaide

- Thesis: Applications of the Octet Baryon Quark-Meson Coupling Model to Hybrid Stars https://doi.org/10.48550/arXiv.1001.4318
- Fortran, MPI, bash

2005 | 2002

Bachelor of Science, Honours
Physics & Theoretical Physics

University of Adelaide

RESEARCH EXPERIENCE

2013 | 2009

Postdoc

Special Research Centre for the Subatomic Structure of Matter

- Dense matter nuclear astrophysics, hybrid stars
- Muonic hydrogen spectroscopy, “the proton radius puzzle”

INDUSTRY EXPERIENCE

Current | 2022

Senior Research Software Engineer

Human Immunology Biosciences (HIBio) & Biogen

- Development of internal tooling supporting translational science, antibody discovery, and infrastructure. Built exploratory dashboards, connected data via APIs, managed deployment of GoAnywhere.
- Migrated contents of a LIMS from Benchling to Sapio. Processing of petabytes of data on UK BioBank.

Current | 2024

Research Software Engineer

Rancho Biosciences / Roche / Genentech

- Integration of AI tooling to shiny dashboard, prompt engineering, data processing in Redshift databases.
- Exploration of ADaM and SDTM data.

Current | 2015

Principal Consultant

Irregularly Scheduled Programming

- Independent contractor / consultant (ABN: 43 486 039 320)
- Selected clients: Gate Biosciences, Lander Analytics

View this CV online with links at jcarroll.com.au/CV

CONTACT

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I have experience working in academia, government, and in both large and small biotech.

LANGUAGE SKILLS

- R
- Rust
- Haskell
- Bash
- Python
- Julia
- APL
- Javascript
- SQL
- Lisp

- English
- Japanese
- French

2022 | 2021

Technical Lead / co-Product Owner

Cancer Immunology Department @ Genentech / Roche

- Coordinated development of a large shiny app project for interactive exploration of genomics visualisation, aimed at life-scientists without coding experience. Remotely directed a team of 10 developers located worldwide.
- R (shiny), CSS, JavaScript, HTML, SQL/SQLite, Docker, Git, CI/CD, database management, statistics, computational biology/genomics, management, agile scrum, leadership

2022 | 2017

Data Curator / Integrator & Software Developer

Cancer Immunology Department @ Genentech / Roche

- Developed software tooling supporting extraction and integration of multiple high-dimensional data modalities for FAIRification of internal data (MultiAssayExperiment / DataMart).
- R (Bioconductor, shiny), CSS, JavaScript, HTML, Docker, Git

2017 | 2013

Senior Research Officer / Fisheries Modeller

South Australian Research and Development Institute (SARDI)

- Developed and maintained fisheries stock assessment models, spatio-temporal catch rate analyses, and geographic information system tools
- Database management, statistics, biology, and the interface between fishers and government policy
- R, C++ (ADMB), SQL, Microsoft Access

COMMUNITY

Current | 2022

Chess Club Teacher (Volunteer)

Highbury Primary School

Current | 2016

Governing Council (Chair, Vice Chair, Subcommittee Chair, General Member)

Highbury Preschool, Highbury Primary, Modbury High

Current | 2016

Editor

rweekly.org

- Compliation, curation, and editing of the rweekly.org newsletter

SELECTED WRITINGS

2025 **Counting Digits Quickly**[1]

2024 **I Patched R to Solve an Exercism Problem**[2]

2024 **Let's Talk About the Weather**[3]

2024 **IPv4 Components in APL**[4]

2023 **Hyperlink Annotations in JavaScript and CSS**[5]

2022 **Australian Signals Directorate 50c Coin Decryption**[6]

2022 **Lissajous Curve Matrix in Julia**[7]

2021 **Adventures in x86 ASM with rx86**[8]

2020 **isEven without modulo**[9]

2018 **Adding strings in R**[10]

I regularly write on my blog https://jcarroll.com.au about puzzles, syntax, and my own learnings as I explore multiple languages - R, Julia, Rust, Python, APL, Haskell, and many more.

SELECTED SOFTWARE PACKAGES

I contribute R packages to open-source software and engage with users, integrating feature requests and bug fixes. I maintain several R packages hosted on CRAN and Bioconductor.

{ntfy} (author, maintainer) CRAN(https://cran.r-project.org/package=ntfy)

- Lightweight Wrapper to the 'ntfy.sh' Service

{ggeasy} (author, maintainer) CRAN(https://cran.r-project.org/package=ggeasy)

- Provides a series of aliases to commonly used but difficult to remember 'ggplot2' sequences

{charcuterie} (author, maintainer) CRAN(https://cran.r-project.org/package=charcuterie)

- Handle Strings as Vectors of Characters

{DFplyr} (author, maintainer) Bioconductor(https://bioconductor.org/packages/release/bioc/html/DFplyr.html)

- A `DataFrame` (`S4Vectors`) backend for `dplyr`

{mathpix} (author, maintainer) CRAN(https://cran.r-project.org/package=mathpix)

- Support for the 'Mathpix' API (Image to 'LaTeX')

{benchlingapi} (author, maintainer) GitHub(https://github.com/jonocarroll/benchlingapi)

- Wraps the Benchling API for access in R

{weatherOz} (contributor) CRAN(https://cran.r-project.org/package=weatherOz)

- An API Client for Australian Weather and Climate Data Resources

{datapasta} (contributor) CRAN(https://cran.r-project.org/package=datapasta)

- R Tools for Data Copy-Pasta

SELECTED PUBLICATIONS

I have experience as a first author and as a collaboator on articles in a diverse range of fields.

2025 **Negative impact of the GABA pathway on αPD-1 PD-L1 immunotherapy**

Submitted

2022 **Natural killer cell educating KIR/HLA combinations impact survival in anti-PD-L1 treated cancer patients**

medRxiv

2021 **Genetic variation associated with thyroid autoimmunity shapes the systemic immune response to PD-1 checkpoint blockade**

Nature Communications, volume 12, number 3355

2020 **Polygenic risk for skin autoimmunity impacts immune checkpoint blockade in bladder cancer**

Proceedings of the National Academy of Sciences, volume 117, number 22, 12288–12294

2018 **Beyond Spreadsheets with R**

https://www.manning.com/books/beyond-spreadsheets-with-r

- Manning Publications
- 352 pages

2016 **Influence of Wind and Swelloncatch Rates in a Dive Fishery: A Case Study from the South Australian Abalone Fishery**

Journal of Shellfish Research, volume 35, number 3, 685-694

2016 **Setting quotas using provisional data: a case study from the South Australian abalone fisheries**

New Zealand Journal of Marine and Freshwater Research, volume 50, number 3, 371-388

2016 **Spawning biomass of jack mackerel (Trachurus declivis) off eastern Australia: Critical knowledge for managing a controversial fishery**

Fisheries Research, volume 179, 10-22

2014 **Quark-meson coupling model, nuclear matter constraints, and neutron star properties**

Physical Review C - Nuclear Physics, volume 89, number 6

2012 **Nuclear quasielastic electron scattering limits nucleon off-mass shell properties**

Physical Review C - Nuclear Physics, volume 86, number 6

2011 **Nonperturbative relativistic calculation of the muonic hydrogen spectrum**

Physical Review A - Atomic, Molecular, and Optical Physics, volume 84, number 1

2011 **Toward a resolution of the proton size puzzle**

Physical Review A - Atomic, Molecular, and Optical Physics, volume 84, number 2

2009 **Phase transition from quark-meson coupling hyperonic matter to deconfined quark matter**

Physical Review C - Nuclear Physics, volume 79, number 4

2005 **Neutral pion decay into $\nu\bar{\nu}$ in dense Skyrmion matter**

Physical Review D - Particles, Fields, Gravitation and Cosmology, volume 71, number 11

LINKS

1: https://jcarroll.com.au/2025/06/29/counting-digits-quickly/
2: https://jcarroll.com.au/2024/02/26/i-patched-r-to-solve-an-exercism-problem/
3: https://jcarroll.com.au/2024/07/27/let-s-talk-about-the-weather/
4: https://jcarroll.com.au/2024/08/22/ipv4-components-in-apl/
5: https://jcarroll.com.au/2023/06/02/hyperlink-annotations-in-javascript-and-css/
6: https://jcarroll.com.au/2022/09/01/asd_coin/
7: https://jcarroll.com.au/2022/05/12/lissajous-curve-matrix-in-julia/
8: https://jcarroll.com.au/2021/12/23/adventures-in-x86-asm/
9: https://jcarroll.com.au/2020/03/09/iseven-without-modulo/
10: https://jcarroll.com.au/2018/10/06/adding-strings-in-r/
11: author, maintainer
12: https://cran.r-project.org/package=ntfy
13: author, maintainer

14: https://cran.r-project.org/package=ggeasy
15: author, maintainer
16: https://cran.r-project.org/package=charcuterie
17: author, maintainer
18: https://bioconductor.org/packages/release/bioc/html/DFplyr.html
19: author, maintainer
20: https://cran.r-project.org/package=mathpix
21: author, maintainer
22: https://github.com/jonocarroll/benchlingapi
23: contributor
24: https://cran.r-project.org/package=weatherOz
25: contributor
26: https://cran.r-project.org/package=datapasta

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